

Laws of Exponents — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1 (a) 2^8 . Same base, add exponents: $3 + 5 = 8$.

Q2 (b) 5^6 . Power of a power: multiply $2 \times 3 = 6$.

Q3 5. Same base, subtract: $9 - 4 = 5$.

Q4 (a) 20^3 . Power of a product: $4^3 \times 5^3 = (4 \times 5)^3$.

Q5 **True**. This is the power-of-a-power law.

Q6 (a) **Both true, R explains A.** $(5^2)^3 = 5^{2 \times 3} = 5^6$.

Short answer

Q7 (i) 3^6 (ii) 6^4 . Add $(4+2)$; subtract $(7-3)$.

Q8 $8^6 = (8^2)^3 = (8^3)^2$. Since $6 = 2 \times 3 = 3 \times 2$.

Q9 $(2^3)^2 = 64$ is larger; $2^3 \times 2^2 = 32$. ($64 > 32$.)

Long answer

Q10 (i) $2^4 (= 16)$ (ii) $3^3 (= 27)$. ($5+3-4 = 4$; $2 \times 4 - 5 = 3$.)

Q11 (i) p^{10} (ii) x^6 . ($4+6 = 10$; $5+3-2 = 6$.)

Q12 Area = $(2^3)^2 = 2^6 = 64 \text{ cm}^2$; Volume = $(2^3)^3 = 2^9 = 512 \text{ cm}^3$.

Total: 21 marks.